Harikrishna S

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.72
CHAVARA PUBLIC SCHOOL, PALA	Class XII (CBSE)	2024	98.20%
BHAVAN'S MUNSHI VIDYASHRAM	Class X (CBSE)	2022	97.20%

WORK EXPERIENCE

Tech & Product Intern | *BuildBay ~ Certificate*

- **Architected** an **end-to-end data pipeline** for smart_estimation_v1, combining CV-based architectural floor-plan preprocessing, **LLM-based building-spec preprocessing**, systematic **graph construction**, and standard-rate-based BoQ generation.
- **Trained** CV models with Detectron2 (ResNet-50 FPN) on an **HPC cluster using SLURM**; fine-tuned YOLOv11 models for ROI detection in architectural plans.
- **Optimized** an **image-processing pipeline** by partitioning processing for problematic datasets and recombining results, **eliminating OOM failures** and simplifying the preceding CV inference setup.

PROJECTS

XV6 Operating System Extension | *Github*

- **Extended MIT xv6** with kernel-level features spanning process scheduling, virtual memory, disk I/O, and software RAID.
- Implemented **12 system calls** for collecting kernel statistics across the implemented features.
- Implemented a **4-level Multi-Level Feedback Queue (MLFQ)** scheduler with periodic priority boosting.
- Built a **demand-paged virtual memory subsystem** with page-fault handling, clock-based page replacement, disk-backed swap, and lazy page loading.
- Developed **pluggable disk I/O schedulers** (FCFS, SSTF) with MLFQ-aware request prioritization.
- Implemented a **software RAID layer supporting RAID 0, RAID 1, and RAID 5** with striping, mirroring, and XOR-based parity recovery.
- **Validated** correctness with **50+ kernel tests**.

Migrator | *Github*

- Developing a **database query migration tool** that automatically adapts optimised SQL queries from a source database to a derived target database by analyzing schema transformations and query semantics.
- Built an **LLM-based test case generation framework** to generate diverse and challenging database workloads using the TPC-H benchmark.
- **Developed** a **baseline query migration system** that transforms source queries into target-compatible queries using source-target mappings and migration rules.

MiniML Language extension | *Github*

- **Extended** MiniML, an **open-source toy language from PLZoo**, with **try-with exception handling and integer division**.
- Extended the MiniML interpreter end-to-end, modifying the lexer, parser, AST, type checker, and evaluator to support exception handling.
- Designed exception propagation and stack unwinding semantics, including control-flow interruption and propagation through nested evaluation.

Query Optimizer Plan-Switch Analysis

- **Analyzed** how PostgreSQL's cost-based optimizer selects execution plans for parametrized **TPC-H** queries, using EXPLAIN ANALYZE and pg_hint_plan to force and compare alternate plans on a 10GB dataset.
- **Designed** an empirical **classification framework** (CORRECT / EARLY / DELAYED) from measured wall-clock execution times to evaluate whether the optimizer's plan choice was well-timed relative to actual runtime crossover points.
- **Found** that a **majority of observed plan switches were mistimed**, surfacing concrete limitations in the optimizer's

cost-based plan selection under parameter sensitivity.

SKILLS

Languages: Python, JavaScript, C++, OCaml

Libraries/Frameworks: PyTorch, OpenCV, FastAPI, React, Pandas

Cloud: AWS (EC2, RDS - PostgreSQL)

Technical Skills: REST API Design, Compiler/Interpreter Design (Lexer, Parser, Type Systems), Operating Systems (Process/Memory Management, Concurrency), Database Management (PostgreSQL), Version Control (Git/GitHub).

RELEVANT COURSES

Mathematics: Probability and Random Variables, Linear Algebra, Introduction to Statistics, Calculus I & II.

Computer Science: Algorithms, Data Structures, Operating Systems I & II, Database Management Systems I & II, Computer Architecture, Compilers I, Formal Methods & Software Verification, Discrete Mathematics